2013-2014 – Master 2 – Macro I

Lecture 3 : The Ramsey Growth Model - Extra Material

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Disclaimer

These are the slides I am using in class. They are not self-contained, do not always constitute original material and do contain some "cut and paste" pieces from various sources that I am not always explicitly referring to (not on purpose but because it takes time). Therefore, they are not intended to be used outside of the course or to be distributed. Thank you for signaling me typos or mistakes at franck.portier@TSE-fr.eu.

6. Comparative dynamics

An unexpected increase in depreciation

- Recall the pair of differential equations that governs the optimal dynamics of the Ramsey model :

$$\frac{\dot{c}_t}{c_t} = \sigma(f'(\frac{k_t}{\bar{L}}) - \delta - \rho) - g.$$

  and

$$\dot{k}_t = \bar{L}f(\frac{k_t}{\bar{L}}) - (\delta + g)k_t - c_t$$

- An increase in δ shifts the locus $\dot{k} = 0$ and the locus $\dot{c} = 0$.

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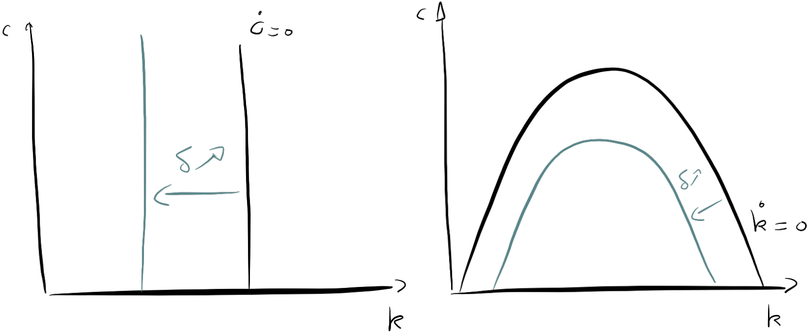
FIGURE 1 : Ramsey with depreciation shock - Phase diagram changes



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- An increase in δ decreases k in the long run, so that consumption is necessarily lower at the new steady state compared to the old one.
- In the short run, depending on the slope of the saddle path , consumption can jump up or down.
- This depends of the relative strength of wealth and substitution effect.
- With high intertemporal elasticity of substitution, c will jump up.
- With low intertemporal elasticity of substitution, c will jump down.

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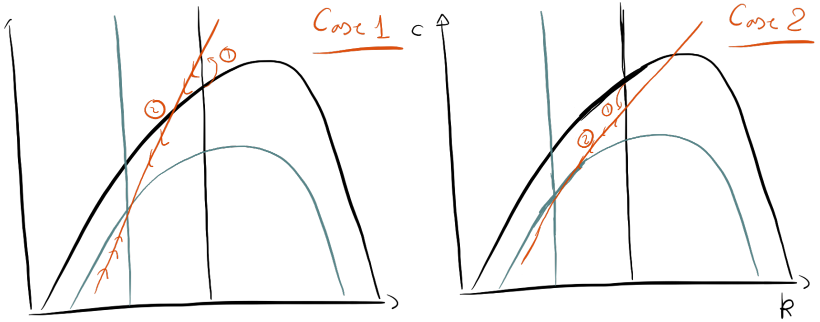
FIGURE 2 : Ramsey with depreciation shock - Possible paths



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- We can go further by looking at a quantitative example.
- Assume time is discrete,
- $u(c) = \frac{c^{1-\sigma}}{1-\sigma}$ with $\sigma = .4$ (high intertemporal substitution) or $\sigma = 4$ (high intertemporal substitution),
- $f(k) = k^{\alpha}$ with $\alpha = .33$.
- Depreciation δ is .025 at the initial steady state, and is increased unexpectedly and permanently in period 1 to the value .05.

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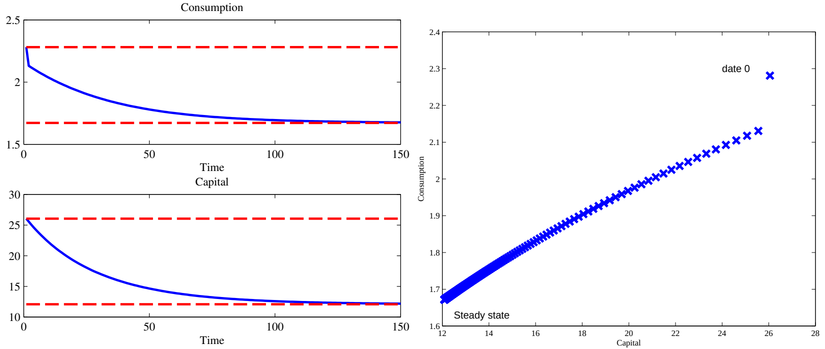
FIGURE 3 : Ramsey with depreciation shock - The case $\sigma = 4$ (low intertemporal substitution) (*Note : On the right panel, each* $\times$ *marker represents couple* (k_t, c_t) *for a given period* t*.*)



6. Comparative dynamics

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FIGURE 4 : Ramsey with depreciation shock - The case $\sigma = .4$ (high intertemporal substitution)(*Note : On the right panel, each* $\times$ *marker represents couple* (k_t, c_t) *for a given period* t.)